

**SIX WEEKS INDUSTRIAL TRAINING
SYNOPSIS**

On

**‘SIGNLEXICON: SIGN LANGUAGE
RECOGNITION SYSTEM USING COMPUTER
VISION AND DEEP LEARNING’**

AT

‘SOLITAIRE INFOSYSTEMS PVT. LTD.’

BY

MANNATDEEP KAUR CHOCHAN

ROLL NO. 28422405159



DEPARTMENT OF MECHANICAL ENGINEERING

GURU NANAK DEV UNIVERSITY, AMRITSAR

‘JULY’ 26’

INTRODUCTION

Artificial Intelligence (AI) and Deep Learning have transformed the field of computer vision by enabling machines to recognize and interpret visual information with high accuracy. One of the significant applications of these technologies is sign language recognition, which aims to bridge the communication gap between deaf or speech-impaired individuals and the general public.

Sign language is a visual language that uses hand gestures, facial expressions, and body movements to convey information. However, many people are unfamiliar with sign language, making communication difficult in educational institutions, workplaces, hospitals, and public environments. To address this challenge, an intelligent recognition system capable of automatically identifying hand gestures can greatly improve accessibility and promote inclusive communication.

The proposed project, **SignLexicon**, is an AI-powered Sign Language Recognition System developed using Computer Vision and Convolutional Neural Networks (CNN). The system recognizes static American Sign Language (ASL) alphabets through uploaded images as well as real-time webcam input. A user-friendly graphical interface has also been developed to make the application simple and accessible for users.

This project demonstrates the practical application of deep learning techniques for gesture recognition while providing a foundation for future research in intelligent human-computer interaction systems.

PROBLEM STATEMENT

Communication between deaf or speech-impaired individuals and people unfamiliar with sign language remains a significant challenge. Traditional communication methods often require the assistance of interpreters, which may not always be available. Existing sign language recognition systems frequently suffer from limited accuracy, dependence on specialized hardware, poor real-time performance, or lack of user-friendly interfaces.

Therefore, there is a need for an efficient, accurate, and easy-to-use sign language recognition system that can recognize hand gestures using ordinary cameras and modern deep learning techniques without requiring expensive hardware.

OBJECTIVES

The major objectives of the proposed system are:

- To develop an AI-based sign language recognition system.
- To recognize static American Sign Language (ASL) alphabets.
- To implement image classification using Convolutional Neural Networks (CNN).
- To provide both image upload and real-time webcam recognition.
- To display predicted sign labels along with confidence scores.
- To develop a simple and interactive graphical user interface using Python Tkinter.
- To promote accessibility and improve communication for hearing and speech-impaired individuals.

LITERATURE REVIEW

Several researchers have proposed sign language recognition systems using Machine Learning, Computer Vision, and Deep Learning techniques. Earlier approaches mainly relied on handcrafted feature extraction methods such as Histogram of Oriented Gradients (HOG), Support Vector Machines (SVM), and Hidden Markov Models (HMM). Although these methods achieved moderate accuracy, they required extensive manual feature engineering and performed poorly under varying lighting conditions and backgrounds.

Recent studies employ Convolutional Neural Networks (CNNs), which automatically learn discriminative features from images and significantly improve recognition accuracy. Many modern systems also integrate OpenCV for image preprocessing and TensorFlow for deep learning model development. However, several existing solutions focus only on offline image recognition or require specialized hardware, limiting their practical usability.

The proposed SignLexicon system addresses these limitations by integrating CNN-based recognition with a user-friendly graphical interface that supports both image upload and real-time webcam prediction using standard hardware.

PROPOSED METHODOLOGY

The proposed SignLexicon system follows a systematic approach for recognizing sign language gestures. Initially, a dataset containing images of American Sign Language alphabets is collected and preprocessed. Image preprocessing includes resizing, normalization, and enhancement to improve model performance.

A Convolutional Neural Network (CNN) is trained using the prepared dataset to learn distinguishing visual features of different hand gestures. After successful training, the trained model is integrated into a Python application developed using the Tkinter library.

The application provides two modes of prediction:

- Image Upload Mode
- Real-Time Webcam Mode

During prediction, the input image is preprocessed and passed to the trained CNN model, which classifies the gesture and displays the predicted alphabet along with the confidence score. OpenCV is used for webcam access and real-time image processing, while TensorFlow performs deep learning inference.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

COMPONENT	SPECIFICATION
Processor	Intel Core i3 or higher
RAM	8 GB Minimum
Storage	10 GB Free Disk Space
Webcam	Integrated/External USB Webcam
Display	1366 × 768 Resolution or Higher

Software Requirements

7.

SOFTWARE	VERSION
Windows Operating System	Windows 10/11
Python	3.x
TensorFlow	Latest Compatible Version
OpenCV	Latest Compatible Version
Keras	Latest Compatible Version
NumPY	Latest Compatible Version
Pillow	Latest Compatible Version
Visual Studio Code	IDE Used

EXPECTED RESULT

The proposed system is expected to accurately recognize static American Sign Language alphabets from both uploaded images and live webcam input. The trained CNN model is capable of providing high classification accuracy while maintaining fast prediction speed. The developed graphical user interface ensures ease of use for users without requiring technical knowledge.

The system demonstrates the practical application of Artificial Intelligence and Computer Vision in improving communication accessibility and serves as a foundation for future intelligent assistive technologies.

FUTURE SCOPE

The proposed system can be further enhanced by incorporating dynamic gesture recognition for complete words and sentences instead of individual alphabets. Future versions may include speech synthesis to convert recognized signs into audible speech, multilingual sign language support, mobile application deployment, cloud-based processing, and integration with Natural Language Processing (NLP) for real-time sentence generation. These improvements can significantly increase the usability and real-world impact of the system.

REFERENCES

- [1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [2] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
- [3] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson Education, 2018.
- [4] A. Kaehler and G. Bradski, *Learning OpenCV 4: Computer Vision with Python*. Sebastopol, CA, USA: O'Reilly Media, 2019.
- [5] F. Chollet, *Deep Learning with Python*, 2nd ed. Manning Publications, 2021.
- [6] TensorFlow Developers, **TensorFlow Documentation**. Available: <https://www.tensorflow.org/>
- [7] OpenCV Team, **OpenCV Documentation**. Available: <https://opencv.org/>
- [8] Python Software Foundation, **Python Documentation**. Available: <https://www.python.org/>
- [9] Pillow Developers, **Pillow Documentation**. Available: <https://python-pillow.org/>
- [10] NumPy Developers, **NumPy Documentation**. Available: <https://numpy.org/>
- [11] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," *arXiv preprint arXiv:1804.02767*, 2018.
- [12] Kaggle, **ASL Alphabet Dataset**. Available: <https://www.kaggle.com/>

